



Round Robin CPU Scheduling Algorithm

Write a C program to implement the Round Robin (RR) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), Burst Times (BT), and a Time Quantum (TQ). The CPU should rotate through the processes in the ready queue, granting each a maximum of one TQ of execution time. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and the overall averages.

Input Format

- 1. The first line of input contains an integer representing the number of processes.
- 2. The second line contains an integer representing the Time Quantum.
- 3. The next lines contain two integers for each process, where

- The first integer denotes the Arrival Time (AT)
- The second integer denotes the Burst Time (BT) of the process.

Output Format

The output displays the Average Turnaround Time and the Average Waiting Time.

Sample Input 1

3
2
0 5
1 3
2 1

Sample Output 1

Enter number of processes:

Enter Time Quantum:

Enter AT and BT for P1:

Enter AT and BT for P2:

Enter AT and BT for P3:

Average Turnaround Time: 6.33

Select Language : C (GCC 9.2.0)

```
1 #include <stdio.h>
2 int main(){
3     int n;
4     printf("Enter number of processes: \n");
5     scanf("%d", &n);
6     int tq;
7     printf("Enter Time Quantum: \n");
8     scanf("%d", &tq);
9     int at[n], bt[n], rt[n];
10    int ct[n], tat[n], wt[n];
11    int i;
12    for(i = 0; i < n; i++){
13        printf("Enter AT and BT for P%d: \n", i+1);
14        scanf("%d %d", &at[i], &bt[i]);
15        rt[i] = bt[i];
16    }
17    int time = 0, comp = 0;
18    while(comp < n){
19        int found = 0;
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2